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5/16/26
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Final Lab

```
In [30]: import sqlite3
import pandas as pd

connection = sqlite3.connect('books.db')
pd.options.display.max_columns = 10
```

Exercise 17.1 a

Select all authors' last names from the authors table in descending order.

```
In [9]: pd.read_sql('SELECT last FROM authors ORDER BY last DESC', connection, index_col=['
```

Out[9]:

last
Wald
Quirk
Deitel
Deitel
Deitel

Exercise 17.1 b

Select all book titles from the titles table in ascending order.

```
In [11]: pd.read_sql('SELECT title FROM titles ORDER BY title', connection, index_col=['titl
```

Out[11]:

title
Android 6 for Programmers
Android How to Program
C How to Program
C++ How to Program
Internet & WWW How to Program
Intro to Python for CS and DS
Java How to Program
Visual Basic 2012 How to Program
Visual C# How to Program
Visual C++ How to Program

Exercise 17.1 c

Use an INNER JOIN to select all the books for a specific author. Include the title, copyright year and ISBN. Order the information alphabetically by title.

```
In [22]: print()
print('Titles by Harvey Deitel:')
print()
pd.read_sql('SELECT t.title, t.copyright, t.isbn FROM titles t INNER JOIN author_IS
connection,
index_col=['title']
)
```

Titles by Harvey Deitel:

Out[22]:

	copyright	isbn
title		
Android 6 for Programmers	2016	0134289366
Android How to Program	2017	0134444302
C How to Program	2016	0133976890
C++ How to Program	2017	0134448235
Internet & WWW How to Program	2012	0132151006
Intro to Python for CS and DS	2020	0135404673
Java How to Program	2018	0134743350
Visual Basic 2012 How to Program	2014	0133406954
Visual C# How to Program	2017	0134601548
Visual C++ How to Program	2008	0136151574

Exercise 17.1 d**Insert a new author into the authors table.**

```
In [26]: cursor = connection.cursor()
cursor = cursor.execute("""INSERT INTO authors (first, last) VALUES ('Bob', 'Smith')
pd.read_sql('SELECT * FROM authors', connection, index_col=['id'])
```

Out[26]:

	first	last
id		
1	Paul	Deitel
2	Harvey	Deitel
3	Abbey	Deitel
4	Dan	Quirk
5	Alexander	Wald
6	Bob	Smith

Exercise 17.1 e**Insert a new title for an author. Remember that the book must have an entry in the author_ISBN table and an entry in the titles table.**

```
In [28]: new_isbn = '0131234567'
author_id = 6
titles_values = (new_isbn, 'The Art of Coding', 1, '2026')
author_isbn_values = (author_id, new_isbn)
```

```

cursor = connection.cursor()
cursor = cursor.execute(
    'INSERT INTO titles (isbn, title, edition, copyright) VALUES (?, ?, ?, ?)',
    titles_values
)
cursor = cursor.execute(
    'INSERT INTO author_ISBN (id, isbn) VALUES (?, ?)',
    author_isbn_values
)

pd.read_sql('SELECT t.title, t.edition, t.copyright, t.isbn FROM titles t INNER JOIN
connection,
params = [author_id],
index_col=['title']
)

```

Out[28]:

	edition	copyright	isbn
title			
The Art of Coding	1	2026	0131234567

Exercise 17.2

Use the description attribute and fetchall() method from the cursor object

In [29]:

```

cursor = connection.cursor()
cursor = cursor.execute('SELECT * FROM titles')
print('description contents')
print(cursor.description)
print()
print('fetchall() contents')
cursor.fetchall()

```

description contents

```
((('isbn', None, None, None, None, None, None), ('title', None, None, None, None, None, None), ('edition', None, None, None, None, None, None), ('copyright', None, None, None, None, None, None)))
```

fetchall() contents

Out[29]:

```
[('0135404673', 'Intro to Python for CS and DS', 1, '2020'),
 ('0132151006', 'Internet & WWW How to Program', 5, '2012'),
 ('0134743350', 'Java How to Program', 11, '2018'),
 ('0133976890', 'C How to Program', 8, '2016'),
 ('0133406954', 'Visual Basic 2012 How to Program', 6, '2014'),
 ('0134601548', 'Visual C# How to Program', 6, '2017'),
 ('0136151574', 'Visual C++ How to Program', 2, '2008'),
 ('0134448235', 'C++ How to Program', 10, '2017'),
 ('0134444302', 'Android How to Program', 3, '2017'),
 ('0134289366', 'Android 6 for Programmers', 3, '2016'),
 ('0131234567', 'The Art of Coding', 1, '2026')]
```

In []: